



University of Stuttgart
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Artificial Intelligence



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Machine Perception and Learning for Collaborative Intelligent Systems

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Transformers

- Embeddings
- Attention Mechanisms
- Attention is all you need
- Bert

Transformers in Computer Vision

Vision Transformer

Object Detection with ViTs

Self-supervised ViTs

Transformers in Computer Vision

Using Transformer in Computer Vision



Source: [Naseer et al., 2021]

- CNNs are hard to solve these cases...



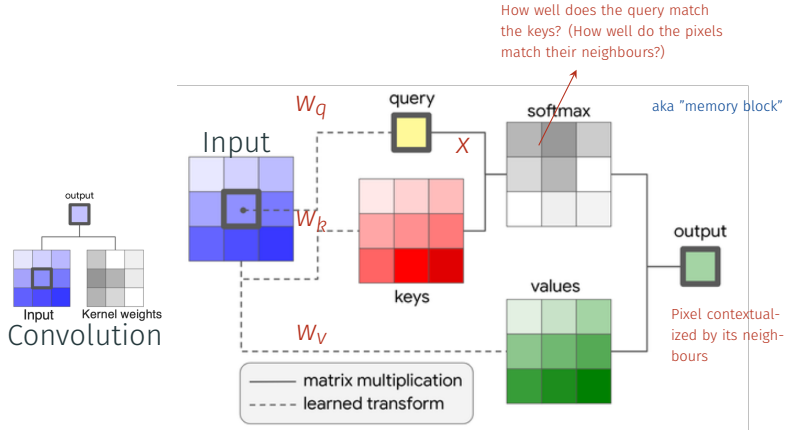
Source: [Naseer et al., 2021]

- CNNs are hard to solve these cases...
- Transformers have better robustness



Source: [Naseer et al., 2021]

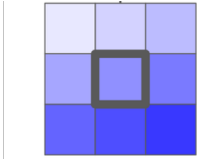
- CNNs are hard to solve these cases...
- Transformers have better robustness
- But also causes trouble for losing positional information



[Ramachandran et al., 2019]

Question: How to encode position information?

Image patch

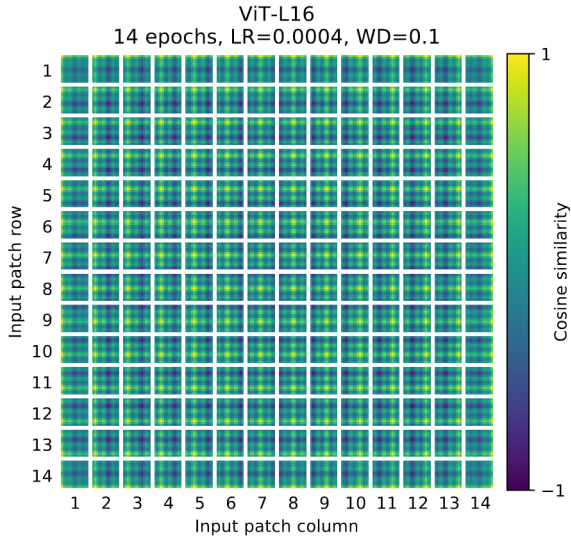


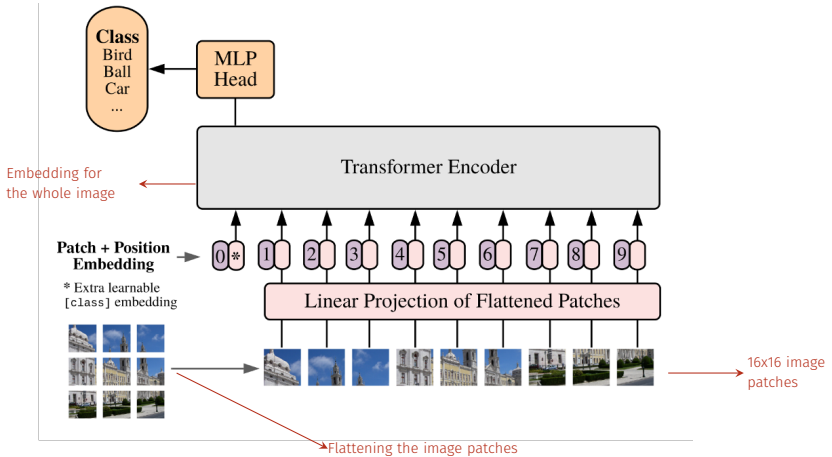
2D position embedding

-1, -1	-1, 0	-1, 1	-1, 2
0, -1	0, 0	0, 1	0, 2
1, -1	1, 0	1, 1	1, 2
2, -1	2, 0	2, 1	2, 2

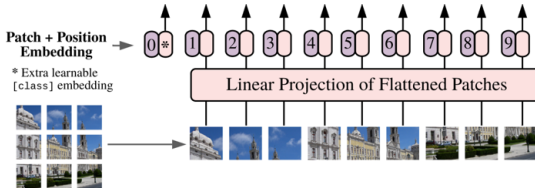
- Relative position embedding (attention) is added to the key

$$y_{ij} = \sum_{a,b \in \mathcal{N}_k(i,j)} \text{softmax}_{ab}(q_{ij}^T k_{ab} + q_{ij}^T r_{a-i,b-j}) v_{ab}$$

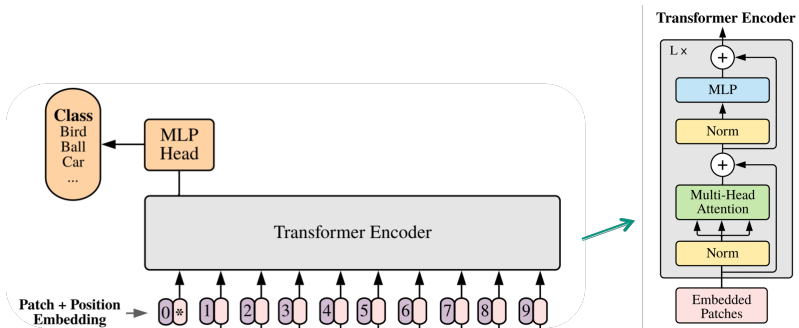




- Image Patch and Position Embedding
 - Fixed-size patches ($16 \times 16 \times 3$) flattened into a vector
 - Linear projection of each patch
 - Additional positional embeddings for [class]



- Encoding and classification
 - Next: standard transformer encoder
 - A classification token is added to the sequence
 - MLP: two dense layers with a GeLU non-linearity



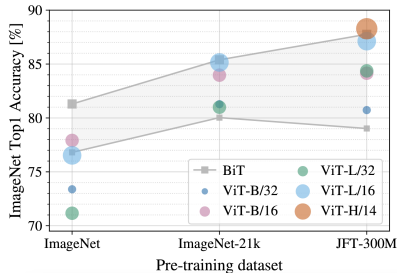
- **Dataset Size Importance:** As Vision Transformers (ViTs) have fewer inductive biases than CNNs, more data is required
- **Dataset Scaling Experiment:** ViT models undergo pretraining on datasets of increasing sizes: ImageNet (1.2M), ImageNet-21k (14.2M), and JFT-300M
- **Performance Optimisation:** To enhance performance on smaller datasets, fine-tune three regularisation parameters: weight decay, dropout, and label smoothing

- Different variants of ViT

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

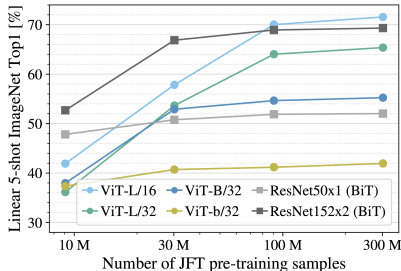
- Pre-training on ImageNet causes ViT-Large to underperform relative to ViT-Base, even with moderate regularization
- With ImageNet-21k pretraining, ViT-Large and ViT-Base show comparable performance
- The full advantage of larger models emerges only with pretraining on JFT-300M

- Large Vision Transformer (ViT) models underperform compared to Big Transfer (BiT) ResNets (indicated by the shaded area) when pre-trained on smaller datasets
- Their performance significantly improves when pre-trained on larger datasets



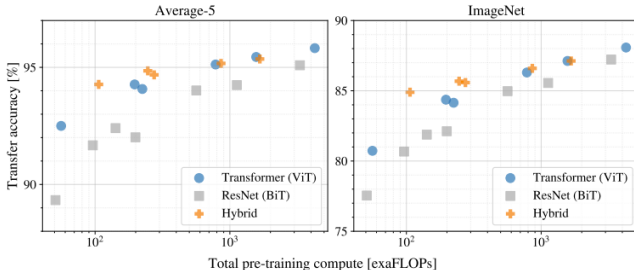
Slide source: [Dosovitskiy et al., 2021]

- Linear Few-shot Evaluation on ImageNet vs. Pre-training Size:
 - ResNets show better performance with smaller pre-training datasets, but reach a performance plateau earlier
 - Vision Transformers (ViT) benefit more from larger pre-training datasets, surpassing the performance of ResNets



Slide source: [Dosovitskiy et al., 2021]

- Performance versus pre-training for different architectures
- Vision Transformers generally outperform ResNets with the same computational budget
- Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models



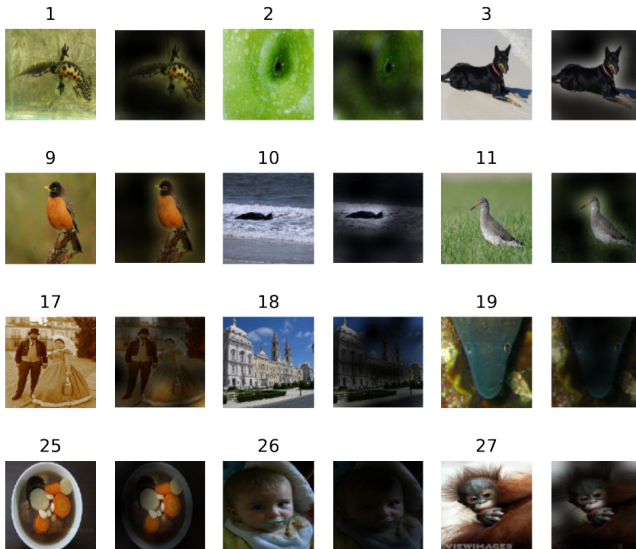
Slide source: [Dosovitskiy et al., 2021]

- ViT models pre-trained on the JFT-300M dataset outperform ResNet-based baselines on all classification datasets
- Huge model + huge data = best performance

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 $\pm$ 0.04	87.76 $\pm$ 0.03	85.30 $\pm$ 0.02	87.54 $\pm$ 0.02	88.4/88.5*
ImageNet Real	90.72 $\pm$ 0.05	90.54 $\pm$ 0.03	88.62 $\pm$ 0.05	90.54	90.55
CIFAR-10	99.50 $\pm$ 0.06	99.42 $\pm$ 0.03	99.15 $\pm$ 0.03	99.37 $\pm$ 0.06	—
CIFAR-100	94.55 $\pm$ 0.04	93.90 $\pm$ 0.05	93.25 $\pm$ 0.05	93.51 $\pm$ 0.08	—
Oxford-IIIT Pets	97.56 $\pm$ 0.03	97.32 $\pm$ 0.11	94.67 $\pm$ 0.15	96.62 $\pm$ 0.23	—
Oxford Flowers-102	99.68 $\pm$ 0.02	99.74 $\pm$ 0.00	99.61 $\pm$ 0.02	99.63 $\pm$ 0.03	—
VTAB (19 tasks)	77.63 $\pm$ 0.23	76.28 $\pm$ 0.46	72.72 $\pm$ 0.21	76.29 $\pm$ 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Source: Metric: accuracy mean and std. dev. (three fine-tuning runs)

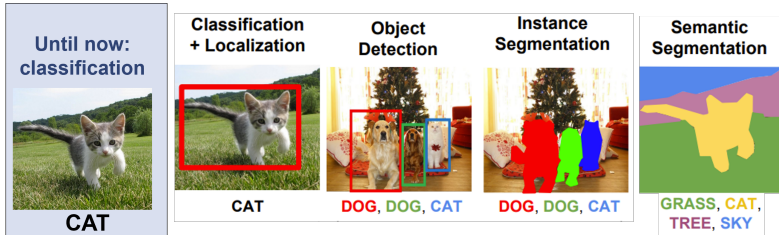
Slide source: [Dosovitskiy et al., 2021]



Transformers in Computer Vision

Object Detection with Vision Transformers

- Localise instances using a bounding box ($x, y, width, height$)
- Classify each bounding box (e.g. *cat*, *tv*)



Classification

Q: what is the object?

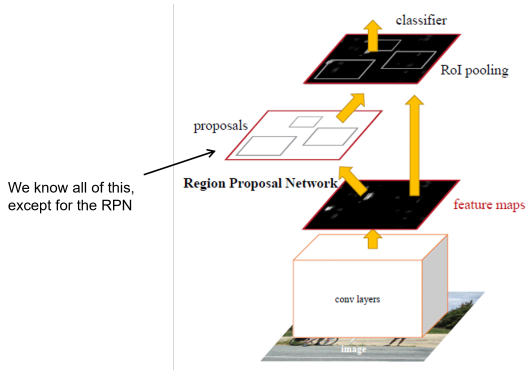
Detection

Q: what and where?

Segmentation

Q: what and where exactly (pixel-level)?

- Example of a CNN-based architecture – FasterRCNN
- Two-stage approach (**Region Proposal Network**)

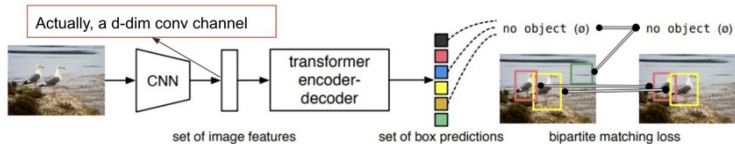


Source: [Ren et al., 2015]

- **Region Proposal Networks (RPN):** In two-stage detectors, an RPN generates potential object bounding boxes (region proposals)
- **ROI Pooling:** Regions of Interest (ROIs) are extracted from feature maps corresponding to the proposed regions
- **Classification and Bounding Box Regression:** Each ROI is classified into object categories, and the BBox coordinates are refined
- **Non-Maximum Suppression (NMS):** Reduces redundant, overlapping boxes by selecting the one with the highest confidence score
- **Multi-Stage Processing:** While the model can be trained end-to-end, components like RPN and NMS introduce multi-stage processing

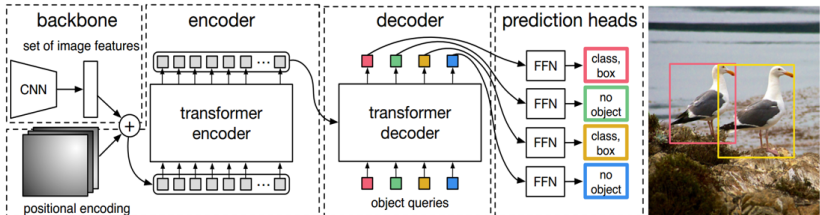
End-to-end object detection with transformers[Carion et al., 2020]

- Object detection (one-to-many) is finally solved by a fully end-to-end approach
- DETR directly predicts the final set of detected objects (no NMS)
- Reduce the hyperparameters to tune, easier for deployment



Source: Leal-Taixé, CS391R

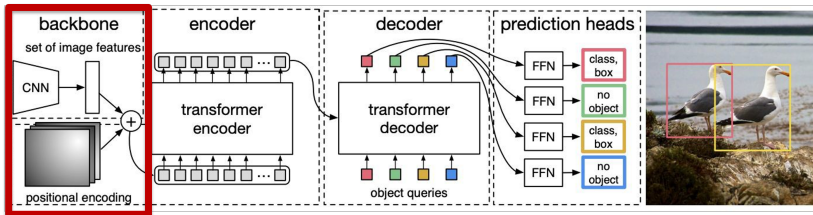
Slide source: [Carion et al., 2020]



Source: Leal-Taixé, CS391R

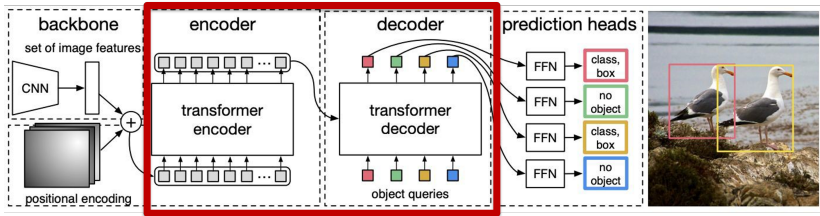
- The CNN is used for feature learning, and the Transformer is used to make predictions
- During training, bipartite matching uniquely assigns predictions with ground truth boxes

Slide source: [Carion et al., 2020]

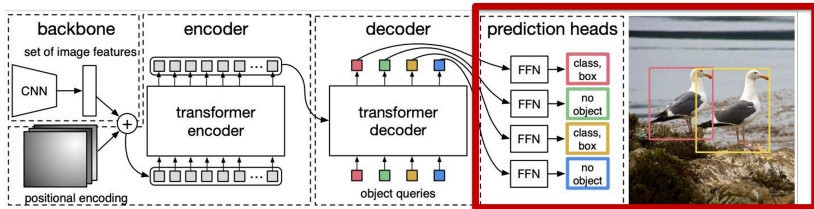


- DETR takes image features extracted from a CNN backbone as input
- the image features are flattened to 1D and supplemented by a positional encoding before passing it into the transformer encoder

Slide source: [Carion et al., 2020]

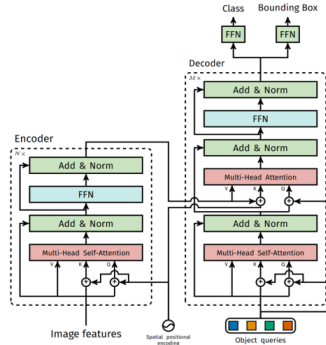


- Encoder: A transformer processes the image features with self-attention and FFN
- Decoder: A transformer decodes the encoder output with fixed number of positional embeddings (object queries)

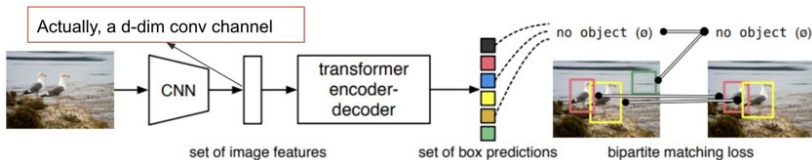


- Each object query output of the decoder is passed to a shared feed-forward network that predicts either a detection (class and bounding box) or a “no object” class

- Very similar to the “Attention is All You Need” architecture, with just a few modifications for object detection
- What are the modifications?



- Given an unordered set of N predictions and N ground truths, find the optimal 1-1 matches between predicted and the ground truth objects
- taking the class, position, and size of objects, **minimize the match cost** for:
 - probability of the predicted class for the label class
 - overlaps between the matched bounding boxes



For object detection, DETR considers a padded set of ground truth objects y and seeks an optimal permutation $\hat{\sigma}$:

$$\hat{\sigma} = \arg \min_{\sigma \in S_N} \sum_{i=1}^N L_{\text{match}}(y_i, \hat{y}_{\sigma(i)}) \quad (1)$$

L_{match} is the **pairwise matching cost** of each ground truth with a prediction. The Hungarian algorithm computes $\hat{\sigma}$ efficiently (see Stewart et al. [2016] for more details on the Hungarian algorithm in neural network training).

After matching predictions with ground truth, the Hungarian loss (summed over all matched pairs) is computed as follows:

$$L_{Hungarian}(y, \hat{y}) = \sum_{i=1}^N \left[-\log \hat{p}_{\sigma(i)}(c_i) + 1_{\{c_i \neq \emptyset\}} L_{box}(b_i, \hat{b}_{\sigma(i)}) \right] \quad (2)$$

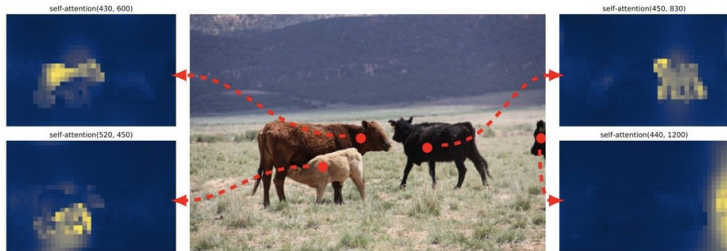
- $\hat{p}_{\sigma(i)}(c_i)$ is the predicted probability for the true class c_i .
- L_{box} represents the loss for the bounding box prediction
- $1_{\{c_i \neq \emptyset\}}$ is an indicator function that is 1 when c_i is not empty (an actual object is present)

AP: Average precision

Model	GFLOPS/FPS	#params	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Faster RCNN-DC5	320/16	166M	39.0	60.5	42.3	21.4	43.5	52.5
Faster RCNN-FPN	180/26	42M	40.2	61.0	43.8	24.2	43.5	52.0
Faster RCNN-R101-FPN	246/20	60M	42.0	62.5	45.9	25.2	45.6	54.6
Faster RCNN-DC5+	320/16	166M	41.1	61.4	44.3	22.9	45.9	55.0
Faster RCNN-FPN+	180/26	42M	42.0	62.1	45.5	26.6	45.4	53.4
Faster RCNN-R101-FPN+	246/20	60M	44.0	63.9	47.8	27.2	48.1	56.0
DETR	86/28	41M	42.0	62.4	44.2	20.5	45.8	61.1
DETR-DC5	187/12	41M	43.3	63.1	45.9	22.5	47.3	61.1
DETR-R101	152/20	60M	43.5	63.8	46.4	21.9	48.0	61.8
DETR-DC5-R101	253/10	60M	44.9	64.7	47.7	23.7	49.5	62.3

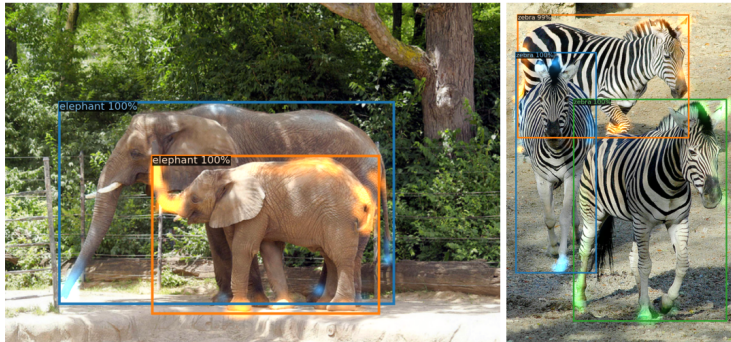
Slide source: [Carion et al., 2020]

The **encoder** is able to separate individual instances



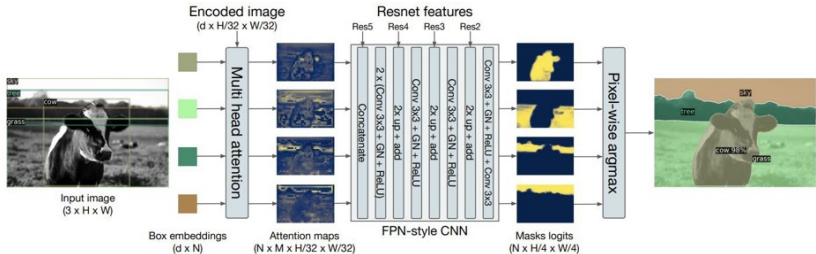
Slide source: [Carion et al., 2020]

Interesting finding: decoder typically attends to object extremities, such as legs and heads.



Slide source: [Carion et al., 2020]

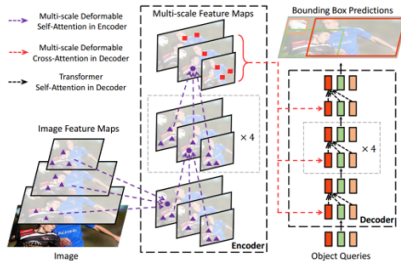
- A binary mask is generated in parallel for each detected object
- then the masks are merged using pixel-wise argmax



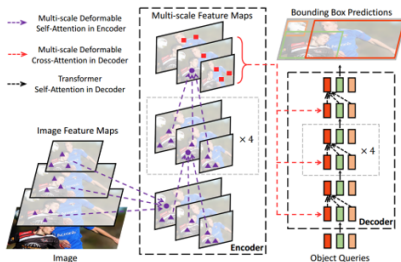
Slide source: [Carion et al., 2020]

- Reaches good results in object detection
- With a minor modification it can be used for panoptic segmentation
- Shortcomings:
 - **Slow convergence** due to initially uniform attention weights, requiring extensive training to focus on relevant locations
 - Ineffective in detecting **small objects**, as attention weights computation scales quadratically with pixel count, complicating high-resolution feature map processing

- **Deformable Attention Module** only attends to a small set of key sampling points (tokens) around a reference point, regardless of the size of the feature maps.



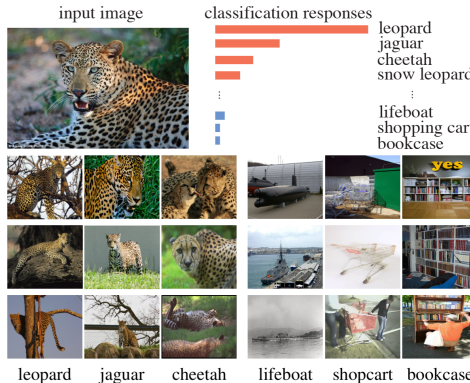
- **Multiscale Deformable Attention** looks over multiple sampling points from multi-scale inputs, instead of only one sampling point.



Transformers in Computer Vision

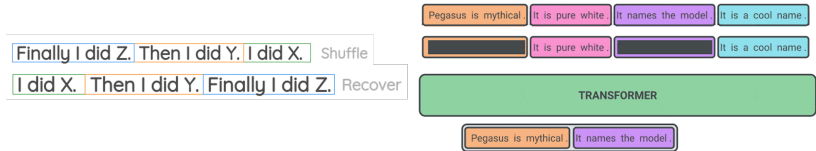
Self-supervised Vision Transformers

- In supervised learning, the apparent similarity in the data themselves that brings some classes closer than others.
- Use self-supervised learning to distinguish similar images



- Transformers have seen tremendous success in NLP
- A key factor in NLP successes was the use of **self-supervised learning**:
 - BERT (Devlin et al., 2019) - next sentence prediction
 - GPT (Radford et al., 2019) - language modelling
- Vision Transformer (ViT) is trained in a fully-supervised manner

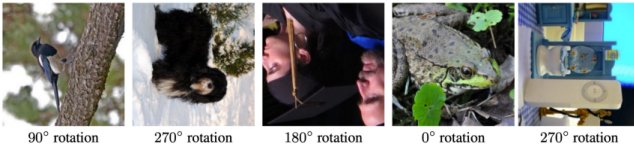
- Predict the order of words in a randomly shuffled sentence
- Predict masked words or sentences in a document (BERT)



Source: Fei-Fei Li, cs231 Lecture

Pretext task: rotation

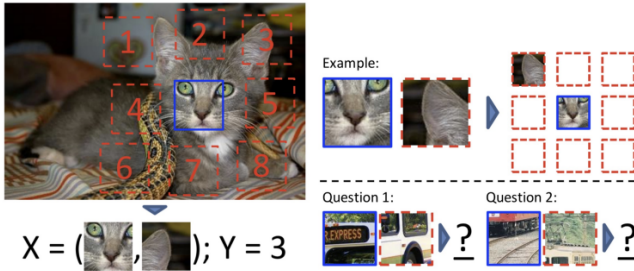
Hypothesis: a model could recognise the correct rotation of an object only if it has the “visual common sense” of what the object should look like unperturbed



Source: Fei-Fei Li, cs231 Lecture

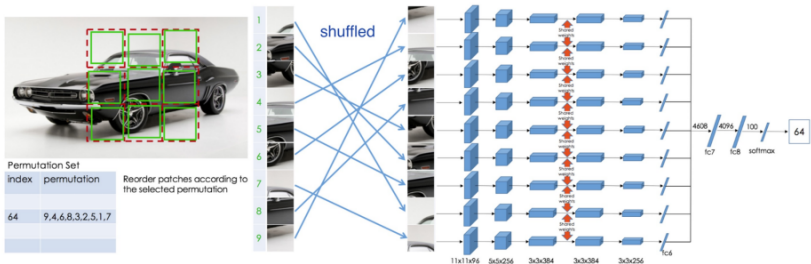
Pretext task: predict relative patch locations

Hypothesis: a model should have “visual common sense” of the relative positions between objects



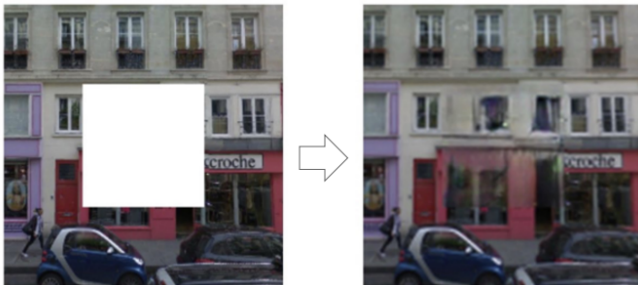
Source: Fei-Fei Li, cs231 Lecture

Pretext task: solving “jigsaw puzzles”



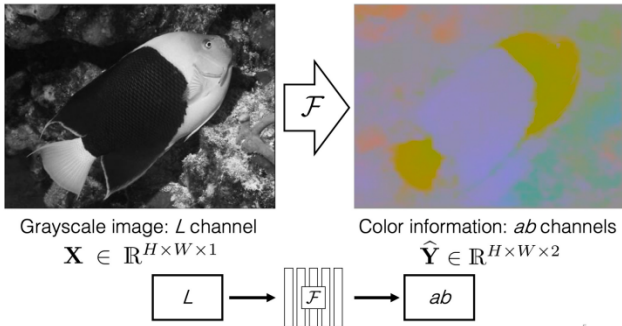
Source: Fei-Fei Li, cs231 Lecture

Pretext task: predict missing pixels (inpainting)



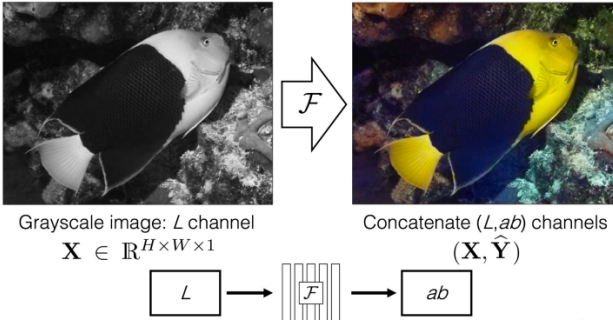
Source: Fei-Fei Li, cs231 Lecture

Pretext task: image colouring



Source: Fei-Fei Li, cs231 Lecture

Pretext task: image colouring



Source: Fei-Fei Li, cs231 Lecture

Problem of solving pretext tasks: learned representations may be tied to a specific task!

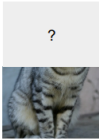


image completion



rotation prediction



"jigsaw puzzle"



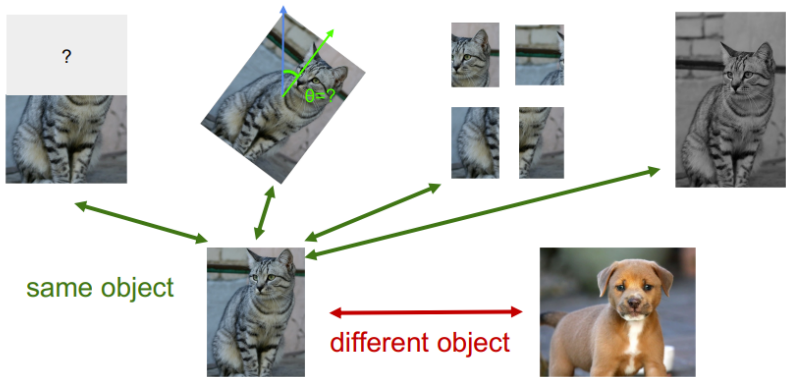
colorization

Source: Fei-Fei Li, cs231 Lecture

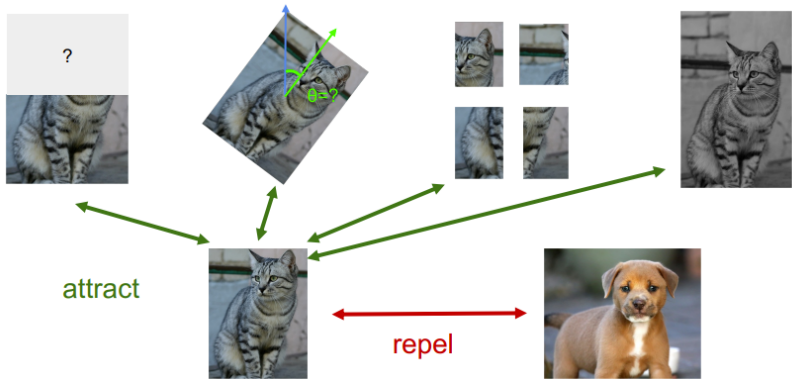
An alternative: **self-supervised contrastive learning**

- Using augmentations, create **different views** of the same image (**positive** pairs)
- Treat other images in the dataset or different augmentations of other images as **negative** pairs
- Minimise the distance between positive examples and maximises the distance to all negative examples

Minimise the distance between positive examples and maximises the distance to all negative examples.



Minimise the distance between positive examples and maximises the distance to all negative examples.



Do different properties emerge from Transformers than CNNs?

- Transformers have different **inductive biases** to CNNs
- No convolutions → no strong principle of **locality**
- It is possible that Transformers behave differently under self-supervision: they may encode **scene layout** or **object boundaries** differently

DINO: DETR with improved denoising anchor boxes for end-to-end object detection [Zhang et al., 2022]

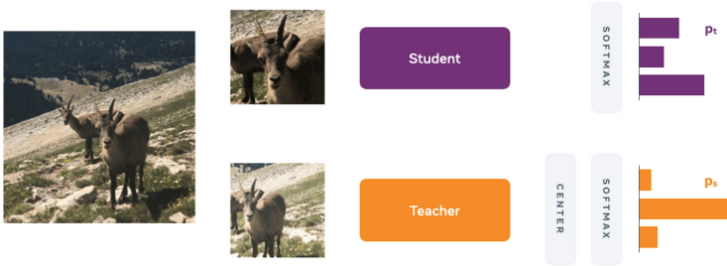
Different Crops of One Image: Uses varied crops to create multiple views

- **Local Views:** Small crops, less than 50% of the image size
- **Global Views:** Large crops, more than 50% of the image size



Local-to-Global Correspondence

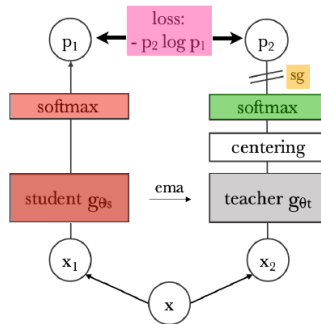
- Encourages the student model to learn context from smaller portions of the image
- Student learns to extrapolate from local views



Slide source: [Zhang et al., 2022]

Source: <https://towardsdatascience.com/paper-explained-dino-emerging-properties-in-self-supervised-vision-transformers-f9386df266f1>

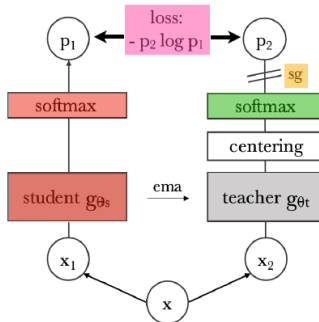
- Makes self-supervised learning into a pseudo classification problem (inspired by [He et al., 2020])
- Aims to learn global representations from local views
- Applied to align the student's distribution with the teacher's



Slide source: [Zhang et al., 2022]

$$P_s(x)^{(i)} = \frac{\exp(\theta_s(x)^{(i)} / \tau_s)}{\sum_{k=1}^K \exp(\theta_s(x)^{(k)} / \tau_s)}$$

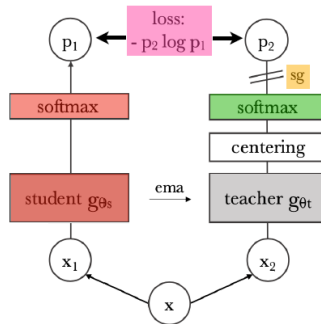
$\tau_s > 0$: a temperature hyper-parameter, θ_s : Parameters of the student model, K : Total number of classes or features



Slide source: [Zhang et al., 2022]

$$P_t(x)^{(i)} = \frac{\exp(\theta_t(x)^{(i)} / \tau_t)}{\sum_{k=1}^K \exp(\theta_t(x)^{(k)} / \tau_t)}$$

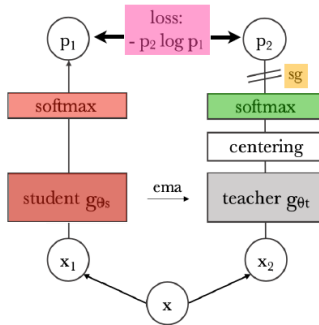
$\tau_t > 0$: another temperature hyper-parameter, θ_t : Parameters of the teacher model, K : Total number of classes or features



Slide source: [Zhang et al., 2022]

$$\min_{\theta_s} H(P_t(x), P_s(x)) \text{ where } H(a, b) = -a \log b$$

Minimised with respect to student parameters.



Source: Samuel Albanie, Foundation Models Online Lectures

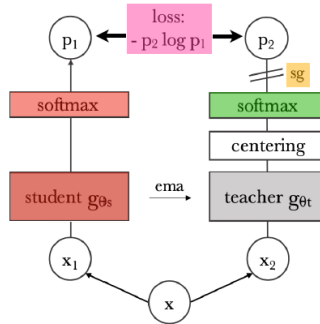
Slide source: [Zhang et al., 2022]

- Loss optimisation for multiple views (crops):

$$\min_{\theta_s} \sum_{x \in \{x_g^1, x_g^2\}} \sum_{x' \in V_x, x'_l} H(P_t(x), P_s(x'))$$

- Only global views are passed to the teacher
- All global and local views (x') are passed to the student
- This encourages local-to-global correspondences

- Mode collapse refers to a situation where the model output is the same across all inputs
- It can manifest in two forms:
 - The output is identical along all dimensions for any input
 - The output is dominated by one dimension
- Centring and sharpening techniques aim to prevent both forms of mode collapse in DINO



Source: Samuel Albanie, Foundation Models Online Lectures

Centering involves adjusting the teacher's raw activations:

$$\text{Logits} = \text{Logits} - \text{Logits_mean}$$

Slide source: [Zhang et al., 2022]

- This process ensures activations are:
 - Positive when above their mean
 - Negative when below their mean
- Prevents any single feature from dominating by setting the mean to the middle of the range
- Softmax assigns low values to negative numbers and high values to positive ones, aiding in balance

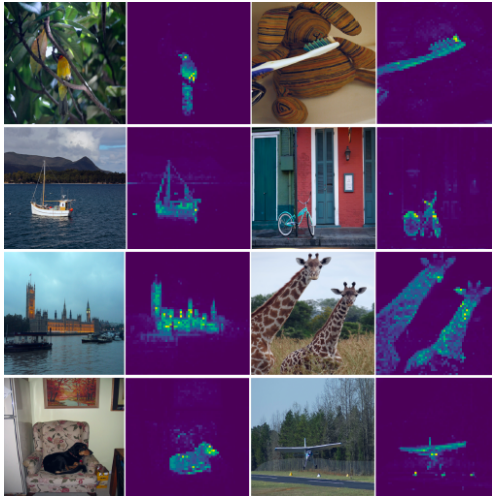
Very effective for transformer backbones!

	Method	Arch.	Param.	im/s	Linear	k-NN
For RN50 DINO is competitive	Supervised	RN50	23	1237	79.3	79.3
	SCLR	RN50	23	1237	69.1	60.7
	MoCov2	RN50	23	1237	71.1	61.9
	InfoMin	RN50	23	1237	73.0	65.3
	BarlowT	RN50	23	1237	73.2	66.0
	OBoW	RN50	23	1237	73.8	61.9
	BYOL	RN50	23	1237	74.4	64.8
	DCv2	RN50	23	1237	75.2	67.1
	SwAV	RN50	23	1237	75.3	65.7
	DINO	RN50	23	1237	75.3	67.5
For ViT-S DINO yields gains	Supervised	ViT-S	21	1007	79.8	79.8
	BYOL*	ViT-S	21	1007	71.4	66.6
	MoCov2*	ViT-S	21	1007	72.7	64.4
	SwAV*	ViT-S	21	1007	73.5	66.3
	DINO	ViT-S	21	1007	77.0	74.5

* baseline re-implemented by DINO authors

Source: Samuel Albanie, Foundation Models Online Lectures

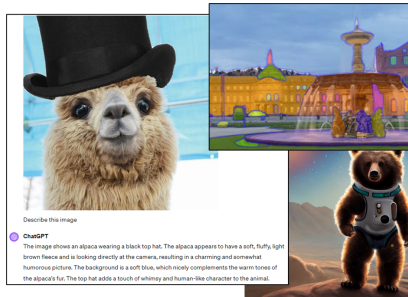
Slide source: [Zhang et al., 2022]



Source: Samuel Albanie, Foundation Models Online Lectures

Slide source: [Zhang et al., 2022]

- LLaMa, Swin, YOLOv10, DINOv2, MoCov3, SAM, etc.
- LLaVA, ViLT, CLIP, BLIP, Mamba, GPT-5, Qwen, etc.
- Stable Diffusion, DALL·E, Sora, WAN, etc.



Multimodal Learning

- Multimodal Representation Learning
- Multimodal Alignment
- Multimodal Transformers
- Multimodal Translation

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